AI-Powered Fire Spread Prediction and Containment System

System Requirements Specification

Panic! At The Kernel
Client: EPI-USE Africa

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1 Introduction

AI-Powered Fire Spread Prediction and Containment System, is designed to address the critical business need for real-time situational awareness and rapid emergency response by utilizing AI-driven simulations and multi-source verification to mitigate the impact of fire disasters. Developed according to IEEE 29148 and ISO 22320 standards, the project's scope encompasses an end-to-end ecosystem that includes secure authentication and role-based access control for users, firefighters, and administrators. The core functionality features a public fire reporting interface with automated verification via NASA FIRMS satellite data, which triggers a sophisticated AI simulation engine to produce color-coded spread predictions based on weather and terrain variables. Furthermore, the system includes a live mapping interface with proximity-based push notifications for registered users, specialized tactical tools for firefighters to log containment lines and view AI recommendations, and an offline map cache to ensure operational continuity in areas with limited connectivity.

2 User Characteristics and User Stories

2.1 User Characteristics

The system supports four user classes, each with distinct access levels and responsibilities.

- **Anonymous Visitor:** Can report fires and view the live map without an account.
- **Registered User:** Receives push notifications and access to AI fire-spread overlays.
- **Firefighter:** Has access to tactical tools, containment line logging, and live field coordinates.
- **Administrator:** Manages user roles, reviews security logs, and maintains the incident archive.

2.2 User Stories

2.2.1 E1: Authentication and Access Control

- As an anonymous visitor, I want to register with an email and password so that I can receive push alerts.
- As a registered user, I want to log in so that I can access my role-based dashboard.
- As an administrator, I want to approve or revoke user roles so that only authorised personnel can access operational tools.

2.2.2 E2: Fire Reporting and Verification

- As any user, I want to report a fire without logging in so that I can alert others immediately.
- As the system, I want to auto-verify reports so that false alerts are minimised.
- As a firefighter or administrator, I want to manually verify or reject pending reports so that unconfirmed incidents receive appropriate review.

2.2.3 E3: AI Fire-Spread Simulation

- As the system, I want to generate a fire-spread prediction upon verification so that users and responders can assess projected risk immediately.
- As the system, I want to re-run the simulation when a containment line is logged so that the prediction reflects current field actions.

2.2.4 E4: Live Fire Map

- As any user, I want to view the live fire map without logging in so that I can see active fires near me.
- As a registered user, I want to see the AI spread-prediction overlay so that I can assess risk in my area.

2.2.5 E5: Notification and Alert System

- As a registered user, I want to receive a push notification when a fire is verified near me so that I can take immediate action.

2.2.6 E6: Firefighter Tactical Tools

- As a firefighter, I want a tactical dashboard with AI recommendations so that I can make informed decisions in the field.
- As a firefighter, I want to draw containment lines on the map so that the AI simulation reflects my ground actions.

2.2.7 E7: Offline Map Cache

- As a firefighter, I want to view cached map data when offline so that I retain situational awareness without connectivity.

2.2.8 E8: Administration Panel

- As an administrator, I want a management panel to approve role requests and revoke access so that system permissions remain controlled.

3 Functional Requirements

3.1 E1: Authentication and Access Control

R1.1 The home screen shall present three options: Log In, Register, and Continue without Account.

R1.2 New registrations shall automatically be assigned the Registered User role.

R1.3 The Administrator role shall require approval by an existing Administrator. The first administrator shall be provisioned by system seed at deployment time.

R1.4 Password reset shall be performed via a time-limited email link. The link shall expire after 15 minutes and be single-use.

R1.5 All role and permission checks shall be enforced server-side at the API layer.

R1.6 JWT tokens shall be used for all authenticated API calls. Access tokens shall be short-lived (e.g., 15 minutes) and paired with refresh tokens.

R1.7 All client-server communication shall use HTTPS/TLS.

R1.8 All authentication events and role changes shall be written to the audit log.

R1.9 Tokens shall be invalidated immediately on logout, role change, or account suspension. Because pure JWTs cannot be instantly revoked without a denylist, the system relies on short TTL and refresh token rotation.

R1.10 Two-factor authentication (TOTP) shall be available for all user roles. Users may enable 2FA after login; a setup endpoint returns an otpauth:// URI.

R1.11 The system shall enforce rate limiting on login attempts. Limit: 5 failed attempts per 15 minutes per account. After 10 consecutive failures, the account shall be locked for 30 minutes.

3.2 E2: Fire Reporting and Verification

R2.1 An unregistered user shall be able to submit a fire report.

R2.2 The fire reporting map shall be accessible from the home screen without login.

R2.3 Required report inputs shall be: map pin location and estimated fire size.

R2.4 Optional report fields shall include: photo attachment and free-text description.

R2.5 The system shall generate a unique reference number for every report.

R2.6 All reports shall enter Pending status; no alerts shall be triggered until Verified.

R2.7 GPS auto-fill shall be offered; manual map pin placement shall always be available.

R2.8 Duplicate reports at the same location shall be automatically linked to one incident.

R2.9 Auto-verification shall trigger on: (a) two independent same-location reports, or (b) a NASA FIRMS satellite match.

R2.10 Manual verification and rejection shall be available to Firefighter and Administrator roles.

R2.11 Rejection shall require a mandatory reason field.

R2.12 Registered reporters shall be notified on verification or rejection of their report.

R2.13 All reports shall be stored in the database for auditing purposes.

R2.14 Report status values shall be: Pending, Verified, and Rejected.

3.3 E3: AI Fire-Spread Simulation

R3.1 The simulation shall start automatically when a fire report is set to Verified.

R3.2 The simulation engine shall ingest: wind speed and direction, temperature, humidity, precipitation, and vegetation dryness index.

R3.3 The simulation engine shall account for terrain slope and vegetation type.

R3.4 Output shall be a colour-coded spread probability grid for 1 h, 3 h, 6 h, and 24 h horizons. Grid resolution: 10 m $\times$ 10 m cells. Probability scale: 5 discrete bands (0–20%, 21–40%, 41–60%, 61–80%, 81–100%).

R3.5 The initial prediction shall be delivered within 5 seconds of verification under normal load (one active fire, weather cache warm). If the simulation exceeds 10 seconds, the engine shall fall back to the last valid prediction (if any) and mark it as "degraded"; otherwise, it shall return a wind-biased elliptical buffer around the ignition point.

R3.6 The engine shall re-run automatically when new weather data is received or a containment line is logged.

R3.7 If a data source is unavailable, the engine shall use last known values and shall never produce a blank map. Predictions generated using data older than 1 hour shall include a data_quality flag indicating which inputs were stale. Such predictions shall be visually distinguished in the UI (e.g., a warning badge)

R3.8 The simulation engine shall be versioned. Each prediction shall record the model version (e.g., "wind_ca_v1.0"). Model version and input timestamps shall be stored in the audit log to enable post-hoc reproducibility.

R3.9 The baseline implementation shall be a wind-biased cellular automaton (CA) model. Fire spreads from ignition cell to neighbouring cells with probabilities proportional to wind speed and direction, reduced by humidity and vegetation dryness. A convolutional neural network (CNN) with LSTM layers for temporal weather trends is a stretch goal, not required for the first demonstrator.

R3.10 The system shall meet a minimum accuracy target evaluated against historical fire incidents. For a held-out test set of at least 20 historical fires (ignition point, weather, and observed final perimeter), the 6-hour predicted spread polygon shall achieve an Intersection-over-Union (IoU) of at least 0.30 against the observed burned area. This metric shall be recomputed after each major model update.

3.4 E4: Live Fire Map

R4.1 The map shall load without login; verified fire markers shall be visible to all users.

R4.2 Spread prediction overlays shall be visible to Registered Users and above.

R4.3 Tapping a fire marker shall display: status, report time, and estimated fire size.

R4.4 The map shall auto-refresh on new data with no manual reload required.

R4.5 The map shall be responsive on mobile, tablet, and desktop.

R4.6 Map tiles shall be cached locally to support basic offline viewing.

3.5 E5: Notification and Alert System

R5.1 Push alerts shall be dispatched within seconds of a fire report reaching Verified status.

R5.2 Alerts shall be delivered as push notifications to registered users.

R5.3 Notifications shall appear when the app is closed or running in the background.

R5.4 Topic-based grid-zone messaging shall be used for simultaneous large-scale delivery.

R5.5 The message broker shall retain and retry failed deliveries.

R5.6 Notification content shall include: fire location, distance from user, severity, and a map deep-link.

R5.7 Every dispatched alert shall be logged with a timestamp, zone ID, and distance.

3.6 E6: Firefighter Tactical Tools

R6.1 Firefighters shall have access to a tactical dashboard displaying a live fire map and AI recommendations.

R6.2 The dashboard shall show: fire direction, high-risk areas, wind speed and direction, and suggested containment locations.

R6.3 The firefighter's GPS position shall be shown as a live icon on the map.

R6.4 Firefighters shall be able to draw containment lines directly on the map.

R6.5 Logged containment lines shall trigger an automatic simulation re-run.

R6.6 Every logged action shall be stored with GPS coordinates and a timestamp.

R6.7 The interface shall be optimised for field use with large tap targets and high contrast.

3.7 E7: Offline Map Cache

R7.1 The app shall cache map tiles and the most recent spread prediction while online.

R7.2 An offline indicator shall be displayed when the device has no connectivity.

R7.3 Cached map data and the most recent prediction shall remain viewable while offline.

3.8 E8: Administration Panel

R8.1 The admin panel shall support: approving and rejecting role requests, searching users, and revoking access.

R8.2 The admin panel shall enforce its own server-side role check.

R8.3 Closed fire incidents shall be moved to a searchable archive automatically.

R8.4 All role changes and access revocations shall be written to the audit log.

4 Use Cases

4.1 Authentication & Access Control

UC-1.1 - User Registration Actor: [3]

- This use case begins with an anonymous visitor tapping register, entering their email and password, and receiving a verification email.
- This use case ends with the visitor confirming their email to receive a registered user role and a JSON Web Token (JWT).

UC-1.2 - User Login Actor: [2]

- This use case begins with a registered user, firefighter, or admin tapping log in and entering their email and password on the login view.
- This use case ends with the user being directed to their role-appropriate home view after the system successfully validates their credentials and issues a JWT.

UC-1.3 - Admin Role Approval / Revocation Actor: [5]

- This use case begins with an admin opening the user management panel to view pending requests or search for an account by name or email.
- This use case ends with the admin approving or rejecting a role request, or revoking account access to manage user permissions.

4.2 Fire Reporting & Verification

UC-2.1 - Fire Report Submission Actor: [4]

- This use case begins with any user tapping on the map to report a fire, selecting or adjusting the GPS location pin, and providing the required fire details on the reporting screen.
- This use case ends with the user tapping submit and receiving a unique reference number as the system sets the report status to pending and triggers the auto-verification check (UC-2.2 [4.2]).

UC-2.2 - Auto-Verification: [4]

- This use case begins with the system detecting a new pending report and checking it against independent nearby reports or NASA FIRMS hotspot data for validation.
- This use case ends with the system updating the report status to verified, triggering the AI simulation, and dispatching push alerts once a verification condition is successfully met (UC-5.1 [4.5]).

UC-2.3 - Manual Verification / Rejection: [4]

- This use case begins with a firefighter or an admin selecting a pending fire report to examine its details and attached photos.
- This use case ends with the reviewer tapping verify to trigger downstream actions or tapping reject with a mandatory reason to update the report status and notify the reporter.

4.3 AI Fire Spread Simulation

UC-3.1 - Generate Fire Spread Prediction:

- This use case begins with the system fetching real-time weather metrics, terrain slope, and vegetation data upon the verification of a fire report.
- This use case ends with the system running the simulation and publishing a color-coded probability grid to the live map for the specified hourly horizons.

UC-3.2 - Re-Run After Containment Line:

- This use case begins with the system receiving containment line coordinates when a firefighter logs a barrier on an active verified fire.
- This use case ends with the system re-running the simulation engine with the containment line as a barrier and publishing the updated prediction to the map within seconds.

UC-3.3 - Re-Run on Weather-Update:

- This use case begins with the system detecting a weather data update that affects an active fire zone and automatically re-running the simulation engine.
- This use case ends with the system publishing the new prediction to the live map and logging the update with a timestamp and a weather-update flag.

UC-3.4 - Auto-Play Fire Spread Simulation:

- This use case begins with a firefighter enabling auto-play and selecting RUN, causing the system to simulate spread for every verified fire and immediately begin animated playback from the point of ignition.
- This use case ends with the firefighter having observed the full progression of each fire's predicted spread over the simulated period, with the option to pause or resume at any point.

UC-3.5 - Inspect Simulation at a Selected Tick:

- This use case begins with a firefighter disabling auto-play, or pausing an active playback, and dragging the timeline slider to a specific tick within the simulated period.
- This use case ends with the map and statistics panel displaying the predicted fire extent and burn counts at that exact tick, allowing the firefighter to assess conditions at a chosen point in time without watching the full animation.

4.4 Live Fire Map

UC-4.1 - View Live Fire Map: [7]

- This use case begins with any user opening the app to immediately load the map and view verified fire markers, with the option to tap a marker for detailed fire status and metrics.
- This use case ends with the user viewing the map as it auto-refreshes with new data, including spread overlays and at-risk zones if the user holds a registered role.

4.5 Notification & Alert System

UC-5.1 - Dispatch Proximity Push Alerts:

- This use case begins with the system identifying at-risk registered users via grid-based topic messaging and dispatching a push notification via AWS notifications within 5 seconds of a fire status changing to verified.
- This use case ends with the user tapping the notification to open the app directly to the fire location on the map as the system logs the alert details, zone ID, and timestamp.

4.6 Firefighter Tactical Tools

UC-6.1 - Tactical Dashboard:

- This use case begins with a firefighter opening the app during an active incident to view the tactical dashboard containing the live fire map, spread overlay, and their own live GPS position.
- This use case ends with the firefighter reviewing the AI panel for critical environmental metrics, high-risk areas, and suggested containment spots to assist with incident monitoring.

UC-6.2 - Log Containment Line:

- This use case begins with a firefighter selecting the draw containment line tool and drawing a line on the map after the device loses connectivity.
- This use case ends with the system saving the coordinates locally with a timestamp and displaying the updated, locally re-run spread prediction incorporating the line as a barrier.

4.7 Offline Map Cache

UC-7.1 - View Map While Offline:

- This use case begins with the app detecting a loss of connectivity and displaying an offline banner while keeping cached map tiles and the most recent spread prediction visible to the firefighter or user.
- This use case ends with the app silently refreshing to live data as it detects a stable network reconnection to acknowledge that the system has resumed normal operations.

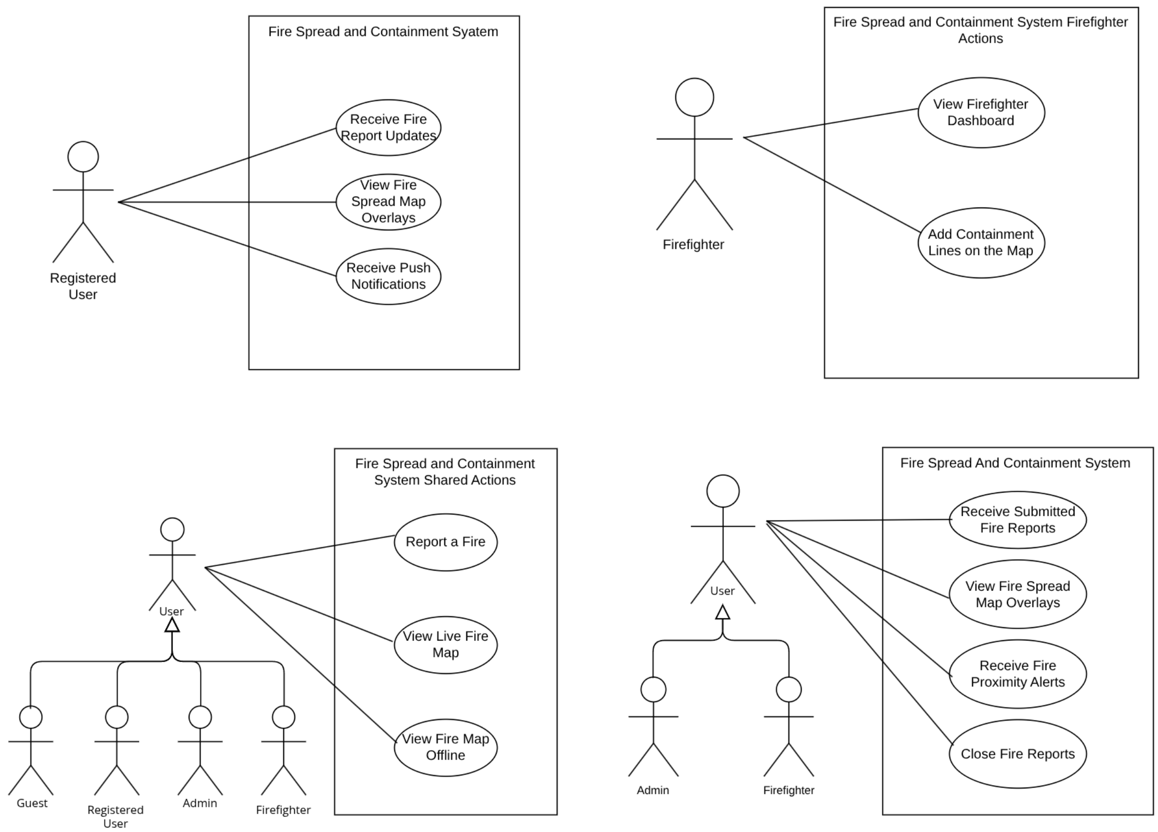

Figure 1: Role Based Access Control

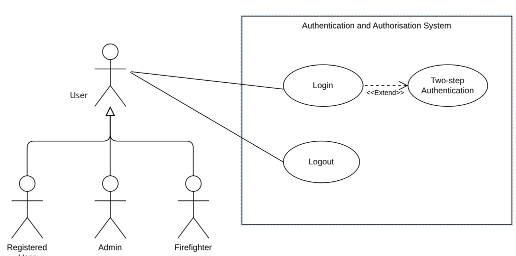

Figure 2: Authentication and Authorisation System

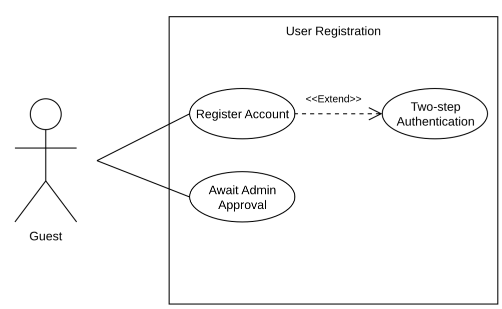

Figure 3: Registration System

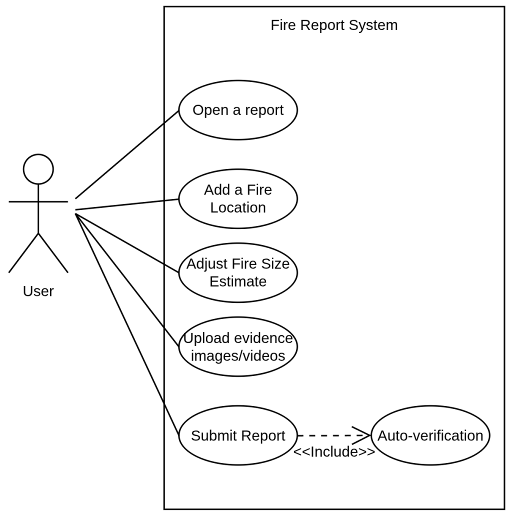

Figure 4: Reporting a Fire System

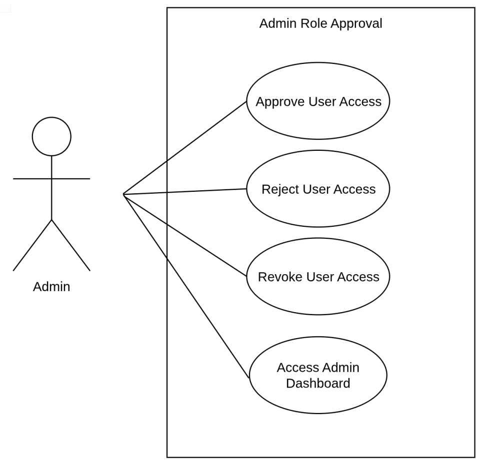

Figure 5: Admin Approval System

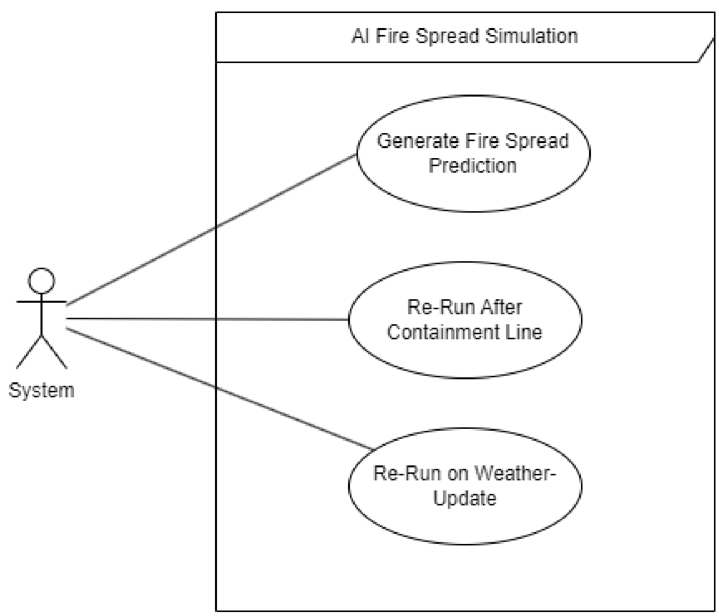

Figure 6: AI Fire Spread simulation

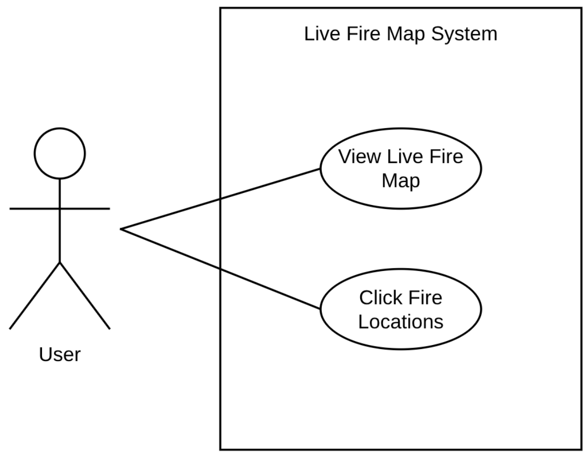

Figure 7: Live Fire Map System

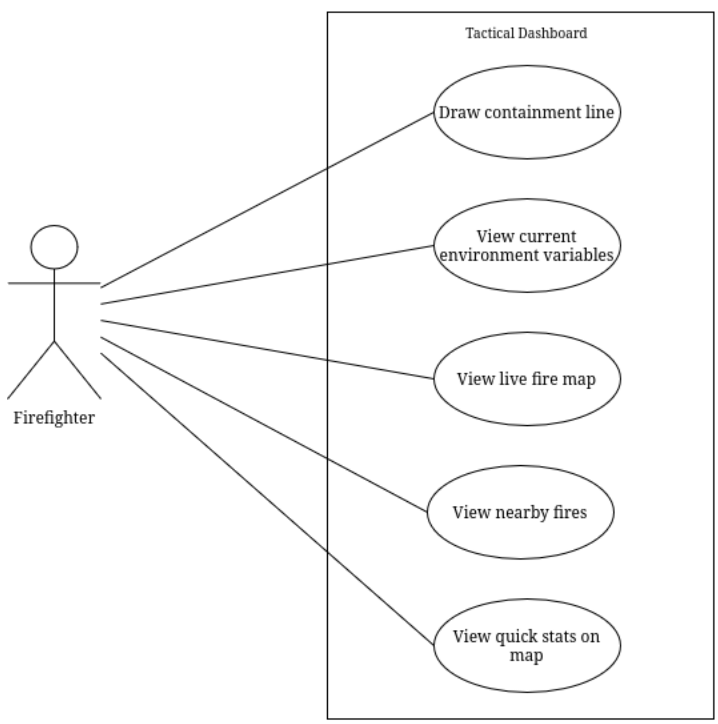

Figure 8: Firefighter Tactical Dashboard System

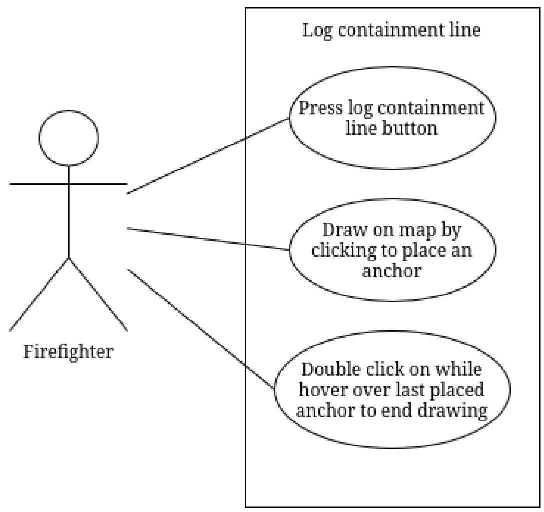

Figure 9: Log Containment Line

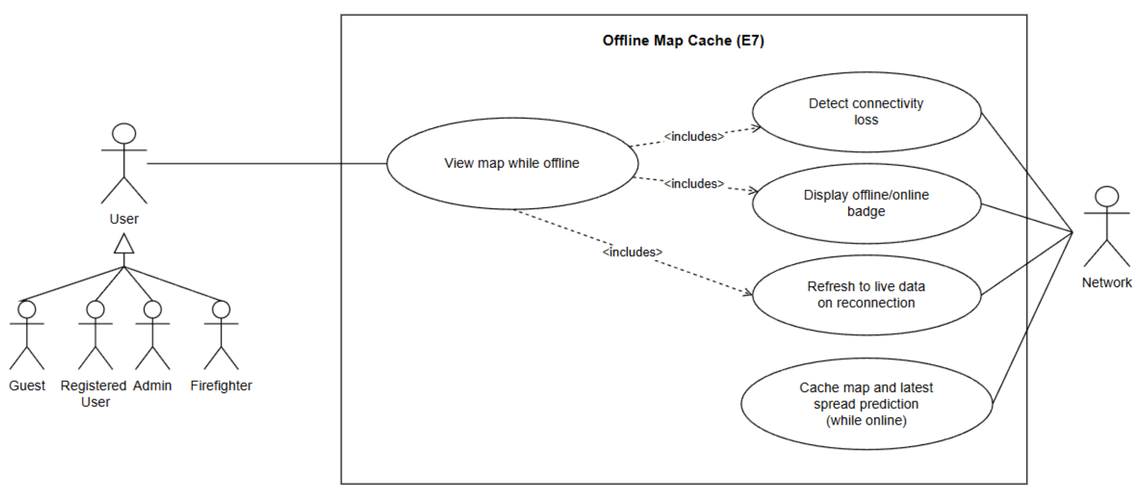

Figure 10: Offline Map Cache System

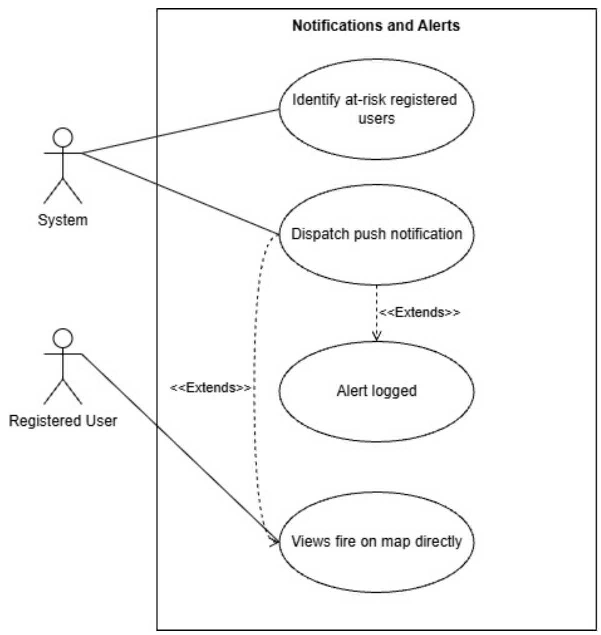

Figure 11: Notification and Alert System

5 Non-Functional Requirements

5.1 Scalability

- Support horizontal scaling to handle a minimum of 1000 concurrent users and fire reports, ensuring smooth simulation and map updates without downtime.

5.2 Security

- End-to-end encryption AES-256 standards is used for user data, role-based access control (RBAC) for firefighters and admins, and secure API authentication for external data sources.

5.3 Availability

- The web and mobile platforms must maintain 99.9% availability, ensuring that real-time updates and interactive maps function optimally across all hardware—including low-spec mobile devices. Furthermore, the system must handle areas with intermittent connectivity by allowing mobile clients to cache critical map sectors locally and queue up to 10 offline incident reports for automatic synchronization upon network reconnection.

5.4 Performance

- Optimized simulation engine and data processing for rapid fire spread predictions, with smart caching for frequently accessed environmental and historical data. The system should calculate and render a standard fire spread prediction within 3 seconds for 95% of simulation requests

5.5 Reliability

- The system must successfully log 100% of user and firefighter containment actions, maintain an AI model prediction accuracy within acceptable standard margins even when up to 25% of environmental data variables are missing, and recover automatically from primary data-stream pipeline disruptions within 2 minutes.

5.6 Maintainability

- Allows integration of additional environmental datasets, satellite sources, or fire risk models must require fewer than 12 person-hours of engineering configuration, while keeping the modular system architecture completely unchanged.

6 Domain Model

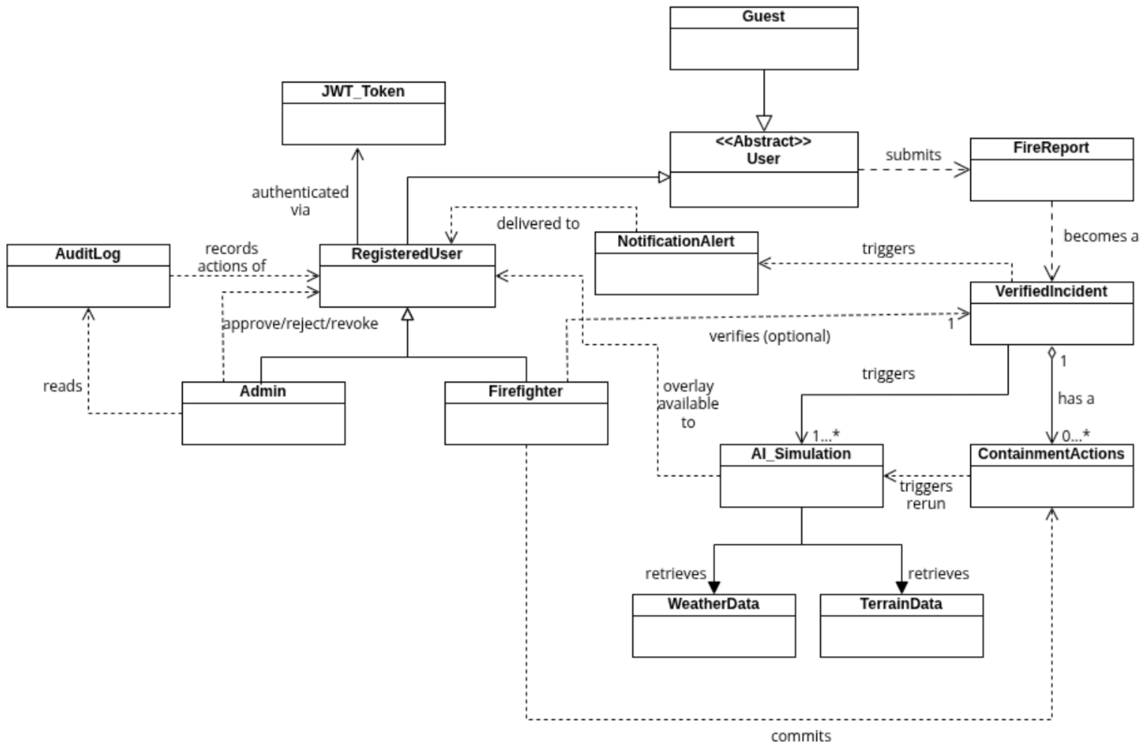

Figure 12: AI-Powered Fire Spread Prediction and Containment System Domain Model

7 Change Log

Date	Version	Change
May	1.0	Initial version
July	2.0	Use Cases, NFR